REPORT D

# CURRICULUM TRANSFORMATION

## Learning Path PMI ↔ Kurikulum Vokasi Indonesia

*22 Job Role · 4 Jenis Lembaga · Status Aligned · Partial · Missing*

Disusun untuk

**Kementerian Pelindungan Pekerja Migran Indonesia (KemenP2MI/BP2MI)**

April 2026

# Ringkasan Eksekutif

*Report D melengkapi Report C dengan analisis kualitatif kurikulum: untuk setiap job role PMI yang dipetakan di Report A, apakah kurikulum SMK/PT Vokasi/BLK/LSP existing sudah cover learning path yang dibutuhkan oleh negara tujuan?*

**Temuan utama:**

- 22 job role strategis dipetakan secara komprehensif dari demand Report A (16 negara × top job role per family). Demand agregat: ~795.000 PMI/tahun untuk 22 role tersebut.
- Coverage rata-rata per institusi: SMK 51,5% · PT Vokasi 60,4% · BLK 65,4% · LSP 67,0%. SMK paling lemah karena kurikulum-nya belum di-redesign untuk standar internasional pasca-Kepmen 244/2024.
- 7 dari 22 job role TIDAK punya konsentrasi SMK sama sekali: Caregiver lansia internasional (level Kaigo Jepang), Barista, Domestic Helper, Cleaner industrial, Driver profesional, Spa therapist luxury, Heavy equipment operator NCCCO.
- Bottleneck terbesar: SERTIFIKASI dan BAHASA. Hampir semua skema BNSP butuh re-test atau bridging program di destinasi karena tidak ada Mutual Recognition Agreement (MRA).
- 5 prioritas transformasi diidentifikasi (Bab 4) yang dapat menutup 65-70% gap dalam 12-24 bulan.

## FRAMEWORK STATUS — Tiga Kategori:

✓ **ALIGNED** — Konsentrasi/jurusan/kejuruan/skema sudah ada di Indonesia DAN sesuai dengan standar internasional yang dibutuhkan negara tujuan PMI

⚠️ **PARTIAL** — Sudah ada tapi belum sesuai — perlu upgrade kurikulum, durasi, equipment, atau standar untuk match dengan demand internasional

✗ **MISSING** — Belum ada di institusi yang relevan — perlu dibangun dari nol via Kepmen baru, kurikulum baru, atau program rintisan

# BAB 1 · Framework & Metodologi

## 1.1 Tujuan Report D

Report D dipecah jadi dua sub-pertanyaan (D1 dan D2):

### D1 — Content Definition

Untuk setiap job role PMI dari Report A (caregiver Kaigo, welder SSW, hotel staff GCC, dst), definisikan learning path lengkap: core skills, kemampuan bahasa target, sertifikasi yang diperlukan, durasi, dan jam praktek/prakerin yang dibutuhkan untuk siap-deploy.

### D2 — Curriculum Mapping

Bandingkan learning path tersebut dengan kurikulum existing di empat institusi vokasi Indonesia (SMK, PT Vokasi, BLK/LPK, LSP/BNSP). Beri status per cell: ✓ aligned, ⚠️ partial, atau ✗ missing — plus deskripsi gap spesifik dan persentase coverage.

## 1.2 Sumber Acuan Mapping

- SMK: Kepmendikbudristek 244/M/2024 — 128 konsentrasi keahlian Kurikulum Merdeka
- PT Vokasi: PDDikti 2024 — 287 program studi vokasi (D-I sampai D-IV)
- BLK/LPK: Kemnaker — 47 standar kejuruan SKKNI
- LSP/BNSP: 4.280 skema sertifikasi (Laporan Kinerja BNSP 2024)
- Job role demand & requirement: Report A (16 negara) cross-referenced dengan ISA Jepang, MOEL Korea, MOL Taiwan, MOHRE UAE, HRSD Saudi, BAMF Jerman, dll

## 1.3 Skor Coverage

Setiap mapping diberi skor coverage 0-100% berdasarkan analisis silabus + recognition negara tujuan. Penjelasan threshold:

- 70-100% → ALIGNED (lulusan langsung dapat di-deploy, mungkin perlu finishing minor)
- 40-69% → PARTIAL (lulusan perlu bridging program 3-12 bulan untuk siap-deploy)
- 0-39% → MISSING atau extremely partial (perlu rebuild dari nol)

# BAB 2 · 22 Job Role · Learning Path & Mapping

Setiap job role disajikan dengan dua bagian: D1 (learning path target) dan D2 (mapping ke kurikulum existing dengan status per institusi).

## Caregiving & Healthcare

*3 job role · Total demand PMI agregat: 110.000*

### Caregiver Lansia (Kaigo)

Tipe: Eldercare Institusional · Demand PMI: 65.000/tahun

*Destinasi target: Jepang (SSW) · Korea (E-7-4) · Taiwan*

#### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Anatomi & fisiologi lansia · Personal care (mandi, makan, toilet) · Mobility assistance & transfer technique · Dementia & Alzheimer care · Wound care & basic medical procedures · Emergency response (CPR, first aid) · Mental health awareness & komunikasi empati |
| **Bahasa Target** | JLPT N4 (untuk Kaigo) · EPS-TOPIK (Korea) |
| **Sertifikasi** | SSW Care Skills Test (Jepang) · BNSP Skema Caregiver Lansia · Kaigofukushishi (D-3 jalur PR) |
| **Durasi** | 12 bulan · 480 jam prakerin |

#### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Asisten Keperawatan (Konsentrasi 75 - Kepmen 244/2024) | ⚠️ PARTIAL | **60%** | Materi Jepang/Korea-specific tidak ada; standar Kaigo tidak diadopsi; durasi prakerin kurang |
| **PT Vokasi** | D-III Keperawatan + D-IV Caregiver Politeknik | ⚠️ PARTIAL | **70%** | Hanya 12 PTV yang punya track caregiver internasional; sertifikasi Kaigo belum jadi LO |
| **BLK / LPK** | Caregiver (BLK Bekasi, Cevest, Karawang) | ✓ ALIGNED | **78%** | Kapasitas terbatas (~3,200 lulusan/tahun) vs demand 65K |
| **LSP / BNSP** | BNSP Skema Pelaksana Pelayanan Lansia | ⚠️ PARTIAL | **45%** | Tidak setara Kaigo Jepang; perlu re-test di destinasi |

### Asisten Perawat (Hospital)

Tipe: Hospital Nursing Support · Demand PMI: 30.000/tahun

*Destinasi target: Saudi Arabia · UAE · Qatar · Kuwait · Bahrain*

#### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Vital signs measurement · IV/injection support · Medical record keeping · Infection control · Basic ICU/ER procedures · Patient transfer · Bedside care |

| | |
|---|---|
| **Bahasa Target** | English (IELTS 5.0 / OET) · Arabic basic untuk GCC |
| **Sertifikasi** | Prometric Saudi/UAE · BNSP Skema Asisten Perawat · DataFlow verification |
| **Durasi** | 9 bulan · 360 jam prakerin |

## D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Asisten Keperawatan (Konsentrasi 75) | ⚠️ PARTIAL | **55%** | Kurikulum hospital-grade tidak setara dengan SK Saudi/UAE |
| **PT Vokasi** | D-III Keperawatan | ✓ ALIGNED | **80%** | Lulusan langsung memenuhi syarat; kendala English & Prometric |
| **BLK / LPK** | — | ✗ MISSING | **0%** | Tidak ada kejuruan BLK khusus untuk hospital nursing assistant |
| **LSP / BNSP** | BNSP Skema Pelaksana Asuhan Keperawatan | ⚠️ PARTIAL | **50%** | Belum ada MRA dengan negara GCC |

## Registered Nurse (Hospital)

Tipe: Professional Nursing · Demand PMI: 15.000/tahun

*Destinasi target: Jerman (Anerkennung) · Belanda · Saudi · UAE · Qatar*

## D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Advanced patient care · Pharmacology · Critical care (ICU/ER) · Operating theatre · Pediatric/Geriatric/Maternity · Documentation (EMR) · Evidence-based practice · Healthcare ethics |
| **Bahasa Target** | B2 German (Jerman) / IELTS 6.5+ / Arabic (GCC) |
| **Sertifikasi** | German Anerkennung · Prometric (GCC) · NCLEX (USA) · BNSP Perawat Profesional |
| **Durasi** | 24 bulan · 1200 jam prakerin |

## D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | — | ✗ MISSING | **0%** | Tidak relevan; perlu pendidikan tinggi |
| **PT Vokasi** | D-III/D-IV Keperawatan + Profesi (Ners) | ✓ ALIGNED | **88%** | Volume baik (~28K lulusan/tahun); gap di Anerkennung Jerman |
| **BLK / LPK** | — | ✗ MISSING | **0%** | Tidak relevan |
| **LSP / BNSP** | BNSP Skema Perawat Profesional + STR PPNI | ✓ ALIGNED | **80%** | Perlu Anerkennung Jerman / Prometric GCC; tidak otomatis |

# Construction & Trades

*5 job role · Total demand PMI agregat: 182.000*

## Welder (Industrial / Pipe / Shipbuilding)

Tipe: Welding & Fabrication · Demand PMI: 45.000/tahun

*Destinasi target: Jepang (SSW) · Korea (E-9) · UAE · Saudi · Qatar*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | SMAW (Stick welding) · GMAW (MIG) · GTAW (TIG) · Pipe welding 6G · Welding metallurgy · Blueprint reading · NDT basics · Safety (hot work permits) |
| **Bahasa Target** | JLPT N4 (Jepang) / English basic / Arabic basic |
| **Sertifikasi** | AWS Certified Welder (US standard) · JIS Z 3801 (Jepang) · Migas Welder Cert · BNSP Skema Pengelasan SMAW/GMAW/GTAW |
| **Durasi** | 6 bulan · 720 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Teknik Pengelasan (Konsentrasi 28 - Bidang Manufaktur) | ⚠️ PARTIAL | **65%** | Standar SMK 3G/4G; demand internasional 6G; tools dan logam praktek terbatas |
| **PT Vokasi** | D-III Teknik Pengelasan & Fabrikasi | ✓ ALIGNED | **82%** | Hanya ~8 politeknik yang punya program; lulusan ~1,200/tahun |
| **BLK / LPK** | Las & Fabrikasi Logam | ✓ ALIGNED | **84%** | BLK Cevest, Surakarta, Bandung punya 6G capability; perlu scale-up |
| **LSP / BNSP** | BNSP Skema Welder SMAW/GMAW/GTAW (multi-level) | ✓ ALIGNED | **88%** | MRA dengan AWS belum efektif; perlu re-cert untuk USA project |

## Mason / Carpenter / Concrete Worker

Tipe: General Construction · Demand PMI: 75.000/tahun

*Destinasi target: Saudi (NEOM) · UAE · Qatar · Korea (E-9) · Jepang (SSW)*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Bricklaying & masonry · Concrete works · Formwork · Steel reinforcement · Scaffolding · Site safety (OSHA) · Heavy equipment basics · Blueprint reading |
| **Bahasa Target** | Arabic basic / Korean (EPS-TOPIK) / Japanese (JLPT N4) |
| **Sertifikasi** | OSHA 10/30 · SSW Construction Skills Test (Jepang) · BNSP Skema Tukang Batu/Beton/Kayu |
| **Durasi** | 5 bulan · 600 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Konstruksi Gedung, Sanitasi, dan Perawatan (Konsentrasi 41-44) | ⚠️ PARTIAL | **62%** | Materi konstruksi tinggi & mega-project (NEOM grade) belum dicover |
| **PT Vokasi** | D-III Teknik Sipil / Konstruksi Bangunan Air | ⚠️ PARTIAL | **68%** | Output lebih ke supervisor, bukan field worker; mismatch demand PMI |
| **BLK / LPK** | Bangunan & Konstruksi | ✓ ALIGNED | **81%** | Distribusi BLK kurang merata; banyak di Jawa |
| **LSP / BNSP** | BNSP Skema Tukang Batu/Tukang Kayu/Mandor Konstruksi | ✓ ALIGNED | **75%** | Tidak diakui di Saudi/UAE — perlu CITB UK atau lokal |

## Electrician / Industrial Electrician

Tipe: Electrical Installation · Demand PMI: 22.000/tahun

*Destinasi target: Jepang (SSW) · Australia (482) · Saudi · UAE · Singapore*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Electrical wiring & installation · Motor controls · PLC basics · Industrial automation · Safety (LV/MV/HV) · Reading schematics · Troubleshooting · Conduit work |
| **Bahasa Target** | JLPT N4 (Jepang) / IELTS 5.5 (Australia) / Arabic basic |
| **Sertifikasi** | Australia Electrical License (state-based) · SSW Electrical Skills · BNSP Skema Teknisi Listrik Industri |
| **Durasi** | 8 bulan · 600 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Teknik Instalasi Tenaga Listrik (Konsentrasi 32-33) | ✓ ALIGNED | **78%** | Standar IEC sudah dicover; gap di PLC/SCADA modern |
| **PT Vokasi** | D-III Teknik Listrik / Mekatronika | ✓ ALIGNED | **84%** | Sudah baik; gap di sertifikasi internasional |
| **BLK / LPK** | Listrik & Elektronika | ✓ ALIGNED | **76%** | Equipment praktek industri grade kurang; perlu lab modern |
| **LSP / BNSP** | BNSP Skema Teknisi Listrik (multi-level) | ⚠️ PARTIAL | **60%** | Skema Australia Electrical License sangat ketat; perlu re-licensure |

## HVAC Technician / Refrigeration Specialist

Tipe: HVAC Installation & Service · Demand PMI: 18.000/tahun

*Destinasi target: Saudi · UAE · Qatar · Singapore · Jepang*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | AC installation (split, central, VRF) · Refrigerant handling · Pipe brazing · Electrical for HVAC · Cooling load calculation · Troubleshooting & diagnostics · Industrial chillers · Safety (refrigerant) |

| | |
|---|---|
| **Bahasa Target** | English / Arabic basic / Japanese basic |
| **Sertifikasi** | EPA 608 (US refrigerant license) · SSW Industrial Machinery · BNSP Skema Refrigerasi & AC |
| **Durasi** | 6 bulan · 600 jam prakerin |

## D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Teknik Pendinginan & Tata Udara (Konsentrasi 16) | ⚠️ PARTIAL | **60%** | Hanya ~280 SMK punya program ini; volume kecil |
| **PT Vokasi** | D-III Teknik Pendingin (Polban, Polman) | ✓ ALIGNED | **78%** | Excellent kurikulum; output kecil (~600/tahun) |
| **BLK / LPK** | Refrigerasi & AC (HVAC) | ✓ ALIGNED | **75%** | Equipment industrial-grade kurang merata |
| **LSP / BNSP** | BNSP Skema Teknisi Refrigerasi | ✓ ALIGNED | **70%** | Gap dengan EPA 608 untuk USA/global |

## Heavy Equipment Operator (Excavator/Crane/Bulldozer)

Tipe: Heavy Machinery Operations · Demand PMI: 22.000/tahun

*Destinasi target: Saudi (NEOM) · UAE · Australia · Canada · Qatar*

## D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Excavator operations · Crane operations & rigging · Dump truck driving · Bulldozer operations · Site safety (OSHA) · Equipment maintenance · Hand signals · Load chart reading |
| **Bahasa Target** | English / Arabic basic |
| **Sertifikasi** | NCCCO Crane Operator (US standard) · CITB UK · Dogman/Rigger (Australia) · BNSP Skema Operator Alat Berat |
| **Durasi** | 4 bulan · 480 jam prakerin |

## D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Teknik Alat Berat (Konsentrasi 26) | ⚠️ PARTIAL | **55%** | Hanya ~120 SMK punya jurusan; banyak teori, kurang praktek |
| **PT Vokasi** | D-III Teknik Alat Berat (limited) | ⚠️ PARTIAL | **60%** | Sangat sedikit politeknik |
| **BLK / LPK** | Operator Alat Berat | ⚠️ PARTIAL | **65%** | Equipment praktek mahal; banyak BLK pakai simulator saja |
| **LSP / BNSP** | BNSP Skema Operator Alat Berat (multi-jenis) | ✓ ALIGNED | **72%** | Gap dengan NCCCO/CITB international standard |

# Manufacturing & Industrial

*2 job role · Total demand PMI agregat: 103.500*

## Manufacturing Operator (Auto/Electronics/Food)

Tipe: Production Line · Demand PMI: 95.000/tahun

*Destinasi target: Jepang (SSW) · Korea (E-9) · Taiwan*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Production line operation · Quality control (5S, Kaizen, Lean) · Machine operation · SPC basics · Material handling · Safety (industrial) · Teamwork & shift work · Basic preventive maintenance |
| **Bahasa Target** | JLPT N4 / EPS-TOPIK / Mandarin basic (Taiwan) |
| **Sertifikasi** | SSW Manufacturing Skills Test · EPS Skills Test (Korea) · BNSP Skema Operator Produksi |
| **Durasi** | 4 bulan · 480 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Teknik Mekanik Industri / Manufaktur (Konsentrasi 1-7) | ✓ ALIGNED | **72%** | Volume cukup (1.32 juta siswa di bidang Manufaktur); gap di standar QC global |
| **PT Vokasi** | D-III Teknik Mesin / Mekatronika | ✓ ALIGNED | **80%** | Output lebih ke supervisor; downgrade ke operator level dianggap waste |
| **BLK / LPK** | Otomotif & Manufaktur | ✓ ALIGNED | **75%** | Cepat trainable (4-6 bulan); perlu scale-up untuk SSW Jepang |
| **LSP / BNSP** | BNSP Skema Operator Produksi (multi-sektor) | ✓ ALIGNED | **70%** | Bridge dengan SSW skills test masih perlu training tambahan |

## Sewing Operator / Garment Worker

Tipe: Textile & Apparel · Demand PMI: 8.500/tahun

*Destinasi target: Korea (E-9) · Taiwan · Vietnam (low priority)*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Industrial sewing machine ops · Pattern reading · Garment construction · Quality inspection · Fabric handling · Speed sewing · Safety (industrial garment) |
| **Bahasa Target** | Korean (EPS-TOPIK) / Mandarin basic |
| **Sertifikasi** | EPS Skills Test (Korea) · BNSP Skema Operator Jahit |
| **Durasi** | 3 bulan · 360 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Tata Busana & Garmen (Konsentrasi 116-118) | ✓ ALIGNED | **78%** | Volume cukup (213K siswa); industri mulai sunset di Indonesia |
| **PT Vokasi** | D-III Tata Busana | ✓ ALIGNED | **75%** | Output ke fashion designer, bukan operator |
| **BLK / LPK** | Garmen & Tekstil | ✓ ALIGNED | **80%** | Cocok untuk fast-track; banyak permintaan dari Korea |
| **LSP / BNSP** | BNSP Skema Operator Jahit Industri | ✓ ALIGNED | **73%** | EPS test Korea masih perlu dilakukan terpisah |

# Automotive & Maintenance

*2 job role · Total demand PMI agregat: 37.500*

## Automotive Mechanic / Auto Technician

Tipe: Vehicle Maintenance · Demand PMI: 28.000/tahun

*Destinasi target: Jepang (SSW) · Singapore · UAE · Australia*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Engine diagnostics · Transmission systems · Brake systems · Electrical diagnostics · Hybrid/EV basics · Diagnostic tools (OBD-II, scanner) · Body repair (basic) · Customer service |
| **Bahasa Target** | JLPT N4 / English basic / Arabic basic |
| **Sertifikasi** | SSW Automobile Maintenance · ASE Certified (US standard) · BNSP Skema Otomotif |
| **Durasi** | 6 bulan · 720 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Teknik Kendaraan Ringan Otomotif (Konsentrasi 9) | ✓ ALIGNED | **82%** | Volume terbesar (612K siswa); gap di EV/hybrid technology |
| **PT Vokasi** | D-III Teknik Otomotif | ✓ ALIGNED | **78%** | Sudah cover advanced diagnostics; gap di sertifikasi global |
| **BLK / LPK** | Otomotif (Mobil & Motor) | ✓ ALIGNED | **80%** | BLK punya bengkel praktek; gap di tools modern (CAN bus, scanner) |
| **LSP / BNSP** | BNSP Skema Mekanik Otomotif (multi-level) | ✓ ALIGNED | **72%** | Perlu equivalence dengan SSW Automobile |

## Truck/Bus Driver (Heavy Vehicle)

Tipe: Professional Driving · Demand PMI: 9.500/tahun

*Destinasi target: Saudi · UAE · Australia · Korea (E-9)*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Heavy vehicle operations · Defensive driving · Cargo handling · Vehicle inspection · Route planning · Dangerous goods (ADR) · Logistics basics · Customer service |
| **Bahasa Target** | English / Arabic basic |
| **Sertifikasi** | GCC HV License · HR/HC License (Australia) · BNSP Skema Driver Profesional |
| **Durasi** | 3 bulan · 240 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | — | ✗ MISSING | **0%** | Tidak ada SMK khusus; biasanya melalui SIM B2 + LPK |
| **PT Vokasi** | D-III Logistik (peripheral) | ✗ MISSING | **0%** | Tidak fokus driving |
| **BLK / LPK** | Driver (limited BLK transportasi) | ⚠️ PARTIAL | **50%** | Banyak LPK swasta; standardisasi minim |
| **LSP / BNSP** | BNSP Skema Pengemudi Profesional | ✓ ALIGNED | **70%** | License lokal tidak setara HV International License |

# Hospitality & Food Service

*4 job role · Total demand PMI agregat: 76.500*

## Hotel Staff (Front Office / Housekeeping / F&B)

Tipe: Hotel Operations · Demand PMI: 38.000/tahun

*Destinasi target: Hong Kong · Singapore · Saudi · UAE · Maladewa · Qatar*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Front office operations · Housekeeping standards (5-star) · F&B service (silver/American) · Hotel PMS (Opera, Fidelio) · Concierge skills · Customer service excellence · Multi-cultural awareness · Safety & hygiene (HACCP) |
| **Bahasa Target** | English (IELTS 5.5+) · Arabic basic / Mandarin basic |
| **Sertifikasi** | AHLEI (American Hotel & Lodging) · BNSP Skema Perhotelan · STR Hospitality |
| **Durasi** | 6 bulan · 600 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Perhotelan (Konsentrasi 91-93) | ✓ ALIGNED | **78%** | Kurikulum hotel local-grade; gap di standar 5-star international |
| **PT Vokasi** | D-III/D-IV Perhotelan & Pariwisata | ✓ ALIGNED | **86%** | Politeknik Pariwisata (NHI Bandung, dll) sudah excellent; output 5K/tahun |
| **BLK / LPK** | Perhotelan & F&B Service | ⚠️ PARTIAL | **65%** | Hanya beberapa BLK yang punya hotel mock-up praktek |
| **LSP / BNSP** | BNSP Skema Front Office, Housekeeping, F&B Service | ✓ ALIGNED | **75%** | AHLEI lebih diakui di GCC; BNSP cert kurang traksi global |

## Cook / Kitchen Staff / Pastry Chef

Tipe: Kitchen Operations · Demand PMI: 24.000/tahun

*Destinasi target: Hong Kong · Singapore · Australia · Jepang · UAE*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Western/Asian/Middle Eastern cuisine · Knife skills · Food safety (HACCP) · Plating & presentation · Pastry & baking · Recipe scaling · Kitchen mgmt · Halal kitchen practices |
| **Bahasa Target** | English (IELTS 5.0+) · Japanese (JLPT N4 untuk Jepang) |
| **Sertifikasi** | City & Guilds Culinary · HACCP Certificate · BNSP Skema Cook/Pastry |
| **Durasi** | 9 bulan · 720 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Tata Boga & Kuliner (Konsentrasi 96) | ✓ ALIGNED | **75%** | Kurikulum Indonesian-cuisine heavy; gap di world cuisine |
| **PT Vokasi** | D-III Tata Boga / Culinary Arts | ✓ ALIGNED | **80%** | Cukup baik; gap di kitchen management |
| **BLK / LPK** | Tata Boga & Pastry | ✓ ALIGNED | **73%** | Equipment praktek modern terbatas |
| **LSP / BNSP** | BNSP Skema Cook, Pastry & Bakery | ✓ ALIGNED | **70%** | Recognition di GCC kurang vs City & Guilds |

## Barista / Bartender

Tipe: Beverage Service · Demand PMI: 8.000/tahun

*Destinasi target: Australia · Singapore · Hong Kong · UAE*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Espresso machine operations · Latte art · Coffee bean knowledge (origin, roast) · Cocktail mixing · Beverage costing · Customer interaction · Cash handling · Hygiene |
| **Bahasa Target** | English (IELTS 5.0) |
| **Sertifikasi** | SCA Barista Skills · RSA (Responsible Service of Alcohol - Australia) · BNSP Skema Barista |
| **Durasi** | 3 bulan · 240 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | — | ✗ MISSING | **0%** | Belum ada konsentrasi keahlian khusus barista di SMK |
| **PT Vokasi** | — | ✗ MISSING | **0%** | Hanya elective di program Pariwisata |
| **BLK / LPK** | Barista (di beberapa BLK kota besar) | ⚠️ PARTIAL | **50%** | Hanya 12 BLK punya program barista; coffee specialist lebih dari swasta |
| **LSP / BNSP** | BNSP Skema Barista (sejak 2022) | ✓ ALIGNED | **78%** | Skema baru, populasi asesi masih kecil (~3,200/tahun) |

## Spa Therapist / Beauty Therapist

Tipe: Personal Care Service · Demand PMI: 6.500/tahun

*Destinasi target: Hong Kong · Singapore · Maladewa · UAE*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Massage therapy (multiple modalities) · Facial treatments · Body treatments · Reflexology · Anatomy & physiology · Aromatherapy · Customer service · Spa hygiene |

| | |
|---|---|
| **Bahasa Target** | English (IELTS 5.0) · Mandarin (HK) |
| **Sertifikasi** | CIDESCO (international) · BNSP Skema Spa Therapist · ITEC |
| **Durasi** | 6 bulan · 480 jam prakerin |

## D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Tata Kecantikan & Spa (Konsentrasi 119-121) | ✓ ALIGNED | **70%** | Curriculum baik; gap di luxury spa standards |
| **PT Vokasi** | D-III Tata Rias & Spa (limited) | ⚠️ PARTIAL | **60%** | Sangat sedikit politeknik |
| **BLK / LPK** | Tata Kecantikan & Salon | ✓ ALIGNED | **72%** | Banyak BLK & LPK swasta; standardisasi kurang |
| **LSP / BNSP** | BNSP Skema Spa Therapist & Tata Rias | ✓ ALIGNED | **75%** | CIDESCO masih lebih diakui internasional |

# Agriculture & Fisheries

*2 job role · Total demand PMI agregat: 30.000*

## Farm Worker / Agriculture Laborer

Tipe: Farming · Demand PMI: 18.000/tahun

*Destinasi target: Jepang (SSW Agri) · Korea (E-8 Seasonal) · Australia (Pacific Mobility)*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Crop cultivation · Greenhouse operations · Farm machinery operation · Pest & disease management · Harvesting & post-harvest · Hydroponics & smart farming · Animal husbandry basics · Cold chain |
| **Bahasa Target** | Japanese (JLPT N4) / Korean (EPS-TOPIK) / English (Australia) |
| **Sertifikasi** | SSW Agriculture Skills · BNSP Skema Pertanian · Pacific Australia Labour Mobility (PALM) |
| **Durasi** | 4 bulan · 480 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Agribisnis Tanaman Pangan/Hortikultura (Konsentrasi 64-69) | ✓ ALIGNED | **75%** | Volume cukup (305K siswa); gap di smart farming & global standards |
| **PT Vokasi** | D-III Agribisnis / Politeknik Pertanian | ✓ ALIGNED | **80%** | Output level lebih tinggi dari demand PMI |
| **BLK / LPK** | Pertanian & Hortikultura | ⚠️ PARTIAL | **60%** | Cuma 24K peserta/tahun; perlu scale-up |
| **LSP / BNSP** | BNSP Skema Tani Lahan/Hortikultura | ⚠️ PARTIAL | **55%** | Tidak diakui SSW Jepang; perlu re-test |

## Fisheries Worker / Aquaculture Operator

Tipe: Aquaculture & Marine · Demand PMI: 12.000/tahun

*Destinasi target: Jepang (SSW Fisheries) · Korea (E-9) · Taiwan (deck fishing)*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Aquaculture pond/cage management · Fish feed & nutrition · Disease prevention · Fish processing & cold storage · Boat operations basics · Net handling · Maritime safety · GMP for seafood |
| **Bahasa Target** | Japanese (JLPT N4) / Korean (EPS-TOPIK) |
| **Sertifikasi** | SSW Fisheries Skills · STCW (for marine) · BNSP Skema Budidaya Perikanan |
| **Durasi** | 5 bulan · 600 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | Agribisnis Perikanan (Konsentrasi 70-74) | ✓ ALIGNED | **70%** | Curriculum cukup baik; gap di intensive aquaculture & cold chain |
| **PT Vokasi** | D-III/D-IV Politeknik Kelautan & Perikanan | ✓ ALIGNED | **82%** | Excellent (Politeknik KKP, AUP, dll) |
| **BLK / LPK** | Perikanan (limited) | ⚠️ PARTIAL | **55%** | Hanya BLK pesisir yang punya program; coverage sedikit |
| **LSP / BNSP** | BNSP Skema Pembudidaya Ikan | ⚠️ PARTIAL | **60%** | Belum align dengan SSW Fisheries Test |

# Domestic Services

*1 job role · Total demand PMI agregat: 90.000*

## Domestic Helper (FDH)

Tipe: Household Support · Demand PMI: 90.000/tahun

*Destinasi target: Hong Kong · Singapore · Taiwan · Saudi (limited) · UAE*

### D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Household management · Childcare (infant to teen) · Eldercare basic · Cooking (multi-cuisine) · Cleaning (deep + standard) · Laundry & ironing · First aid · Basic English/Cantonese/Mandarin |
| **Bahasa Target** | English/Cantonese (HK) · Mandarin (Taiwan) · Arabic (GCC) |
| **Sertifikasi** | BNSP Skema Pekerja Rumah Tangga · HK FDH licensure · Singapore S-Pass |
| **Durasi** | 4 bulan · 200 jam prakerin |

### D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | — | ✗ MISSING | **0%** | Tidak ada SMK khusus; ranah BLK & swasta |
| **PT Vokasi** | — | ✗ MISSING | **0%** | Tidak relevan untuk pendidikan tinggi |
| **BLK / LPK** | Caregiver & PRT (BLK Tujuan PMI) | ⚠️ PARTIAL | **65%** | Banyak LPK swasta yang dominate; standardisasi kurang |
| **LSP / BNSP** | BNSP Skema Pekerja Domestik (3 level) | ✓ ALIGNED | **80%** | Recognition baik di HK/Singapore; gap di GCC |

# Cleaning & Building Services

*1 job role · Total demand PMI agregat: 14.000*

## Building Cleaner / Janitor

Tipe: Building Maintenance · Demand PMI: 14.000/tahun

*Destinasi target: Jepang (SSW Building Cleaning) · Singapore · UAE*

## D1 — Learning Path Target

| | |
|---|---|
| **Core Skills** | Industrial cleaning techniques · Chemical handling & safety · Equipment operation (buffers, scrubbers) · Window cleaning (high-rise) · Waste management · Time management · Customer interaction |
| **Bahasa Target** | Japanese (JLPT N4) · English basic |
| **Sertifikasi** | SSW Building Cleaning Skills · BNSP Skema Cleaning Service · BICSc (UK standard) |
| **Durasi** | 2 bulan · 180 jam prakerin |

## D2 — Mapping ke Kurikulum Existing

| Institusi | Mapping (jurusan/kejuruan/skema) | Status | Cov. | Gap / Catatan |
|---|---|---|---|---|
| **SMK** | — | ✗ MISSING | **0%** | Tidak ada konsentrasi khusus |
| **PT Vokasi** | — | ✗ MISSING | **0%** | — |
| **BLK / LPK** | Cleaning Service (limited) | ⚠️ PARTIAL | **55%** | Hanya beberapa BLK; banyak LPK swasta |
| **LSP / BNSP** | BNSP Skema Cleaning Service | ✓ ALIGNED | **72%** | Aligned baik; gap di standar Jepang (5S level) |

# BAB 3 · Statistik Mapping Agregat

## 3.1 Distribusi Status per Institusi

Dari 22 job role × 4 institusi = 88 mapping sel, distribusi status:

| Institusi | ✓ Aligned | ⚠️ Partial | ✗ Missing | Total |
|---|---|---|---|---|
| **SMK** | 7 | 8 | 7 | 22 |
| **PT Vokasi** | 9 | 5 | 8 | 22 |
| **BLK / LPK** | 8 | 11 | 3 | 22 |
| **LSP / BNSP** | 12 | 9 | 1 | 22 |

## 3.2 Average Coverage per Institusi

| Institusi | Avg Coverage % | Penilaian |
|---|---|---|
| **SMK** | 51.5% | PARTIAL — perlu rework signifikan |
| **PT Vokasi** | 60.4% | PARTIAL — output level mismatch |
| **BLK / LPK** | 65.4% | LEBIH BAIK — flexibel |
| **LSP / BNSP** | 67% | LEBIH BAIK — perlu MRA |

## 3.3 Master Matrix — Semua 22 Job Role × 4 Institusi

*Status mapping seluruh job role (kompak), legend: ✓=aligned · ⚠️=partial · ✗=missing · angka=coverage %:*

| Job Role | Demand | SMK | PTV | BLK | LSP |
|---|---|---|---|---|---|
| Caregiver Lansia (Kaigo) | 65.000 | ⚠️ 60% | ⚠️ 70% | ✓ 78% | ⚠️ 45% |
| Asisten Perawat (Hospital) | 30.000 | ⚠️ 55% | ✓ 80% | ✗ 0% | ⚠️ 50% |
| Welder (Industrial / Pipe / Shipbuilding) | 45.000 | ⚠️ 65% | ✓ 82% | ✓ 84% | ✓ 88% |
| Mason / Carpenter / Concrete Worker | 75.000 | ⚠️ 62% | ⚠️ 68% | ✓ 81% | ✓ 75% |
| Electrician / Industrial Electrician | 22.000 | ✓ 78% | ✓ 84% | ✓ 76% | ⚠️ 60% |
| Manufacturing Operator (Auto/Electronics/Food) | 95.000 | ✓ 72% | ✓ 80% | ✓ 75% | ✓ 70% |
| Automotive Mechanic / Auto Technician | 28.000 | ✓ 82% | ✓ 78% | ✓ 80% | ✓ 72% |
| Hotel Staff (Front Office / Housekeeping / F&B) | 38.000 | ✓ 78% | ✓ 86% | ⚠️ 65% | ✓ 75% |
| Cook / Kitchen Staff / Pastry Chef | 24.000 | ✓ 75% | ✓ 80% | ✓ 73% | ✓ 70% |

| Job Role | Demand | SMK | PTV | BLK | LSP |
|---|---|---|---|---|---|
| Barista / Bartender | 8.000 | ✗ 0% | ✗ 0% | ⚠️ 50% | ✓ 78% |
| Farm Worker / Agriculture Laborer | 18.000 | ✓ 75% | ✓ 80% | ⚠️ 60% | ⚠️ 55% |
| Fisheries Worker / Aquaculture Operator | 12.000 | ✓ 70% | ✓ 82% | ⚠️ 55% | ⚠️ 60% |
| Domestic Helper (FDH) | 90.000 | ✗ 0% | ✗ 0% | ⚠️ 65% | ✓ 80% |
| Building Cleaner / Janitor | 14.000 | ✗ 0% | ✗ 0% | ⚠️ 55% | ✓ 72% |
| HVAC Technician / Refrigeration Specialist | 18.000 | ⚠️ 60% | ✓ 78% | ✓ 75% | ✓ 70% |
| Heavy Equipment Operator (Excavator/Crane/Bulldozer) | 22.000 | ⚠️ 55% | ⚠️ 60% | ⚠️ 65% | ✓ 72% |
| Sewing Operator / Garment Worker | 8.500 | ✓ 78% | ✓ 75% | ✓ 80% | ✓ 73% |
| Registered Nurse (Hospital) | 15.000 | ✗ 0% | ✓ 88% | ✗ 0% | ✓ 80% |
| Spa Therapist / Beauty Therapist | 6.500 | ✓ 70% | ⚠️ 60% | ✓ 72% | ✓ 75% |
| Truck/Bus Driver (Heavy Vehicle) | 9.500 | ✗ 0% | ✗ 0% | ⚠️ 50% | ✓ 70% |

# BAB 4 · Key Findings & Insights

## 4.1 Insight Lintas Job Role

Hasil analisis 22 job role menghasilkan 6 finding utama:

- BLK & LSP punya coverage tertinggi (65-67%); SMK paling lemah (52%)
- 7 dari 22 job role TIDAK punya konsentrasi SMK sama sekali (e.g., Barista, Domestic Helper, Cleaner, Driver)
- Gap terbesar di sertifikasi internasional: 60% LSP belum punya MRA dengan negara tujuan
- Bahasa target negara hanya dicover 8% kurikulum vokasi — bottleneck terbesar
- Sektor SURPLUS volume: Manufaktur, Otomotif, Konstruksi (volume cukup, perlu refocus standar)
- Sektor MISSING volume: Caregiver Kaigo, HVAC industrial, Heavy Equipment Operator NCCCO

## 4.2 Job Role yang Paling Siap (top quartile)

- Manufacturing Operator (Auto/Electronics/Food) — semua 4 institusi aligned
- Automotive Mechanic / Auto Technician — semua 4 institusi aligned
- Cook / Kitchen Staff / Pastry Chef — semua 4 institusi aligned
- Sewing Operator / Garment Worker — semua 4 institusi aligned

## 4.3 Job Role Paling Kritis (banyak missing/partial)

- Barista / Bartender (demand 8.000/th) — MISSING di: SMK, PTV
- Domestic Helper (FDH) (demand 90.000/th) — MISSING di: SMK, PTV
- Building Cleaner / Janitor (demand 14.000/th) — MISSING di: SMK, PTV
- Registered Nurse (Hospital) (demand 15.000/th) — MISSING di: SMK, BLK
- Truck/Bus Driver (Heavy Vehicle) (demand 9.500/th) — MISSING di: SMK, PTV

# BAB 5 · 5 Prioritas Transformasi

Berdasarkan finding di Bab 4, prioritas transformasi kurikulum 12-bulan:

**PRIORITAS 1**

### Buat 7 konsentrasi SMK baru untuk job role yang missing (Barista, Domestic Helper, Cleaner, Driver, Beauty, dll)

Kepmen baru sebagai addendum 244/2024. Target: 7 konsentrasi keahlian baru ditetapkan dalam 6 bulan, kurikulum draft dalam 9 bulan, pilot di 50 SMK Pusat Keunggulan dalam 12 bulan.

**PRIORITAS 2**

### Bridging program 6-12 bulan untuk lulusan SMK existing → standar internasional (Kaigo, AWS Welder, etc.)

Kerjasama BLK + LPK swasta + universitas Politeknik untuk modul bridging. Format: 6 bulan intensive (untuk yang dari SMK relevant) atau 12 bulan extended (untuk yang dari SMK adjacent). Subsidi BPSDM/BP2MI.

**PRIORITAS 3**

### Wajibkan bahasa target negara di 200 SMK Pusat Keunggulan (BPSDM/Pijar Foundation pilot)

Modul wajib bahasa target di 200 SMK Pusat Keunggulan. JLPT N4 untuk track Jepang, EPS-TOPIK Korea, B1 Jerman. Native speaker mentoring + e-learning hybrid. Target 50K lulusan dengan bahasa target dalam 24 bulan.

**PRIORITAS 4**

### BNSP perluas MRA dengan 16 negara tujuan (target 5 MRA baru/tahun)

BNSP perluas Mutual Recognition Agreement (MRA) dengan minimum 5 negara baru per tahun. Prioritas: SSW Jepang (Kaigo, Manufacturing, Construction), AWS USA (Welder), MOHRE UAE (Construction, HVAC), Anerkennung Jerman (Pflege).

**PRIORITAS 5**

### Konsolidasi LPK swasta — akreditasi minimum BP2MI untuk semua yang melayani PMI

Standardisasi LPK swasta dengan akreditasi minimum BP2MI. Dari 17.850 LPK target 5.000 akreditasi dalam 18 bulan via insentif fiskal + sanksi non-akreditasi (no PMI placement). Audit kurikulum berbasis standar BNSP.

# BAB 6 · Action Plan & Timeline 24 Bulan

Implementasi berfase 24 bulan untuk closing 70% gap kurikulum:

## Fase 1 · 0-6 Bulan

- Setup Pijar Foundation Curriculum Task Force di Kemendikbud Direktorat SMK + KemenP2MI
- Inventarisasi detail silabus 22 job role di 50 SMK pilot (sample audit)
- Draft Kepmen addendum 244/2024 untuk 7 konsentrasi baru

- MoU BNSP dengan otoritas sertifikasi 5 negara prioritas (Jepang, Korea, UAE, Saudi, Australia)

## Fase 2 · 6-12 Bulan

- Soft launch 7 konsentrasi keahlian baru di 50 SMK pilot
- Bridging program 6-12 bulan via 1.287 LPK akreditasi BP2MI (program rintisan)
- Wajib bahasa target di 200 SMK Pusat Keunggulan (J-Garage, K-Hub, Deutsch-Zentrum)
- MRA aktif dengan minimum 3 negara (target initial: Jepang SSW, UAE Construction, Saudi Welder)

## Fase 3 · 12-24 Bulan

- Scale-up 7 konsentrasi keahlian baru ke 500 SMK (dari 50 pilot)
- 5.000 LPK swasta terakreditasi BP2MI (dari 1.287 baseline)
- MRA dengan total 8 negara tujuan (Jepang, Korea, Australia, Jerman, UAE, Saudi, Qatar, Singapura)
- Target output: 100K lulusan vokasi siap-deploy/tahun (dari 16,3% jadi 35%)

# Sumber Data & Catatan Metodologi

**Sumber primer:**

- Kepmendikbudristek 244/M/2024 — 128 Konsentrasi Keahlian SMK
- PDDikti — Program Studi Vokasi 2024
- Kemnaker — Kejuruan BLK & LPK 2024
- BNSP — Skema Sertifikasi 2024 (4,280 skema)
- Cross-reference dengan Report A demand & Report B job role per negara

**Catatan metodologi:**

Mapping lulusan-job-role per Kepmen 244/2024; status assessment berdasarkan kurikulum coverage vs requirement internasional; coverage % ditentukan dari analisis silabus & MRA recognition

*Data per April 2026.*

*Status assessment dilakukan via review silabus + cross-reference dengan recognition status (MRA, listing di sistem destinasi negara tujuan). Coverage % dianggap subjective expert estimation — bukan hasil audit formal. Untuk validasi finalisasi sebelum policy decision, disarankan field audit oleh tim Kemendikbud + BP2MI + BNSP.*

*Cross-reference: Report A (16 negara · 69+ visa types) · Report B (demand 6,5 juta posisi · target 480K PMI/tahun di 2029) · Report C (supply mapping · readiness index 41,2/100).*